

$$1) \quad f(x) = \begin{cases} \frac{x^2}{8} & \text{se } 0 \leq x \leq 1 \end{cases} \quad \text{Grupo I}$$

$$P(x < 1) = f(1) = \frac{1}{8}$$

$$2) \quad F_x(c) = \begin{cases} 0 & \text{se } c < 0 \\ \frac{c^2}{8} & \text{se } 0 \leq c \leq 1 \\ \frac{1}{8} + \frac{7}{8}(c-1) & \text{se } 1 \leq c < 2 \\ 1 & \text{se } c \geq 2 \end{cases}$$

$$3) \quad 0$$

$$4) \quad \frac{1}{32}$$

$$5) \quad \frac{1}{2} = \frac{1}{8} + \frac{7}{8}(c-1) \Leftrightarrow c = \frac{10}{7}$$

$$6) \quad \text{Não houve}$$

Grupo II

$$1) \quad P(Y < -4)$$

Ao padronizar

$$z = \frac{-4 - 0}{2} = -2$$

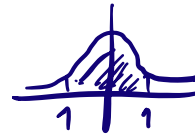
$$P(z < -2) = P(z > 2)$$

$$= P(z < 2) = 0,5 - 0,4772 = 0,0228$$

$$2) P(|Y| \leq 2) \\ = P(-2 \leq Y \leq 2)$$

Padronizando

$$P(-1 \leq Z \leq 1) \\ = 2 \times P(0 \leq Z \leq 1) \\ = 2 \times 0,3413 = 0,6826$$



$$3) E[-2Y] = 0$$

$$\text{Var}[-2Y] = 16$$

$$E[T-2Y] = 3 - 0 = 3$$

$$\text{Var}[T-2Y] = 9 + 16 = 25$$

$$N(3, 25)$$

4) A média da soma nos 100 dias

$$E[S_{100}] = 100 \times E[Y] = 100 \times 0 = 0$$

Variação da soma

$$\text{Var}(S_{100}) = 100 \times \text{Var}(Y) = 100 \times 4 = 400$$

$$\text{Desvio padrão} = \sqrt{400} = 20$$

$$P(S_{100} > 5)$$

padronizando

$$Z = \frac{5 - 0}{20} = \frac{1}{4}$$

$$P(Z > 1/4) = 0,5 - P(0 \leq Z \leq 1/4) = 0,5 - 0,0987 \\ = 0,4013$$

Grupo III

1) Sim

2) Sim

3) $0,4 \times 0,6$

4) $0,4 \times 0,6 + 0,2 \times 0,3 + 0,1 \times 0,1$

5)
$$\frac{0,1 \times 0,1}{0,4 \times 0,6 + 0,2 \times 0,3 + 0,1 \times 0,1}$$

6)
$$\frac{0,3 \times 0,8 + 0,1 \times 0,1}{1 - (0,4 \times 0,6 + 0,2 \times 0,3 + 0,1 \times 0,1)}$$

$1 \approx 0,55$

Grupo IV

1) $\Omega = \{ (x, y, z) : x, y, z \in \{0, 1, 2, 3, 4, 5, 6\} \}$

2) $I = \{ (1, 1, 1), (2, 2, 2), (3, 3, 3), (4, 4, 4), (5, 5, 5), (6, 6, 6) \}$

O acontecimento não é elementar pois o espaço amostral não tem apenas um resultado

3) $P(A \cap B \cap C) = 0$

$P(A) = \frac{1}{2}$; $P(B) = \frac{1}{2}$; $P(C) = \underbrace{\left(\frac{1}{2} \times \frac{1}{2}\right)}_{P_A + P_B} + \underbrace{\left(\frac{1}{2} \times \frac{1}{2}\right)}_{I_{A+B} + I_{A+B}}$

$P(A) \times P(B) \times P(C) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8} \neq 0 = \frac{1}{2}$

4)

5) a - Distribuição Binomial

$$P(X=k) = \binom{3}{k} \cdot \left(\frac{1}{2}\right)^k \cdot \left(\frac{1}{2}\right)^{3-k}$$

$$P(X=k) \begin{cases} 0 & 1 & 2 & 3 \\ \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8} \end{cases}$$

$$b - F_X(c) = \begin{cases} 0 & \text{se } c < 0 \\ \frac{1}{8} & \text{se } 0 \leq c < 1 \\ \frac{1}{2} & \text{se } 1 \leq c < 2 \\ \frac{7}{8} & \text{se } 2 \leq c < 3 \\ 1 & \text{se } c \geq 3 \end{cases}$$

$$c - P(\bar{X} \geq 1 \mid P(X \geq 1))$$

$$\begin{aligned} &= \frac{P(\bar{X} \geq 1 \cap X \geq 1)}{P(X \geq 1)} = \frac{P(\bar{X} = 2) + P(\bar{X} = 1)}{1 - P(X = 0)} \\ &= \frac{\binom{3}{2} \times \left(\frac{1}{2}\right)^2 \times \frac{1}{2} + \binom{3}{1} \times \frac{1}{2} \times \left(\frac{1}{2}\right)^2}{1 - \left(\binom{3}{0} \cdot \left(\frac{1}{2}\right)^3\right)} \\ &= \frac{\frac{3}{8} + \frac{3}{8}}{\frac{7}{8}} = \frac{\frac{6}{8}}{\frac{7}{8}} = \frac{6}{7} \end{aligned}$$

